



TARGET-90

JKBOSE CLASS-11TH

Bhaichara
ON
TOP.

#XUV ♡

CHEMICAL THERMODYNAMICS

One Shot

CHEMISTRY

Vishal Sir



JKBose Wallah

Topics To Be Covered

1 ... One Shot

2 ...

3 ...

4 ...

UNIT-V: THERMODYNAMICS

09 MARKS

- Concepts of system, types of systems, surrounding, work, heat; energy intensive and extensive properties, state functions. First Law of Thermodynamics, internal energy, enthalpy, heat capacity, specific heat, molar heat capacity, measurement of E and H . Hess's law of constant heat summation, enthalpy of bond dissociation, combustion; formation, atomization, sublimation, phase transition ionization and dilution.
- Introduction of entropy as a state function, free energy change for spontaneous and non-spontaneous process and equilibrium.

Some Basic Terms of Thermodynamics



1. System and Surroundings:

The part of the universe we choose to study (temperature, pressure, energy changes, etc.) is called a system.

Everything else outside the system is called the surroundings.

Universe = System + Surroundings

2. Types of Systems:

(a) Open System:

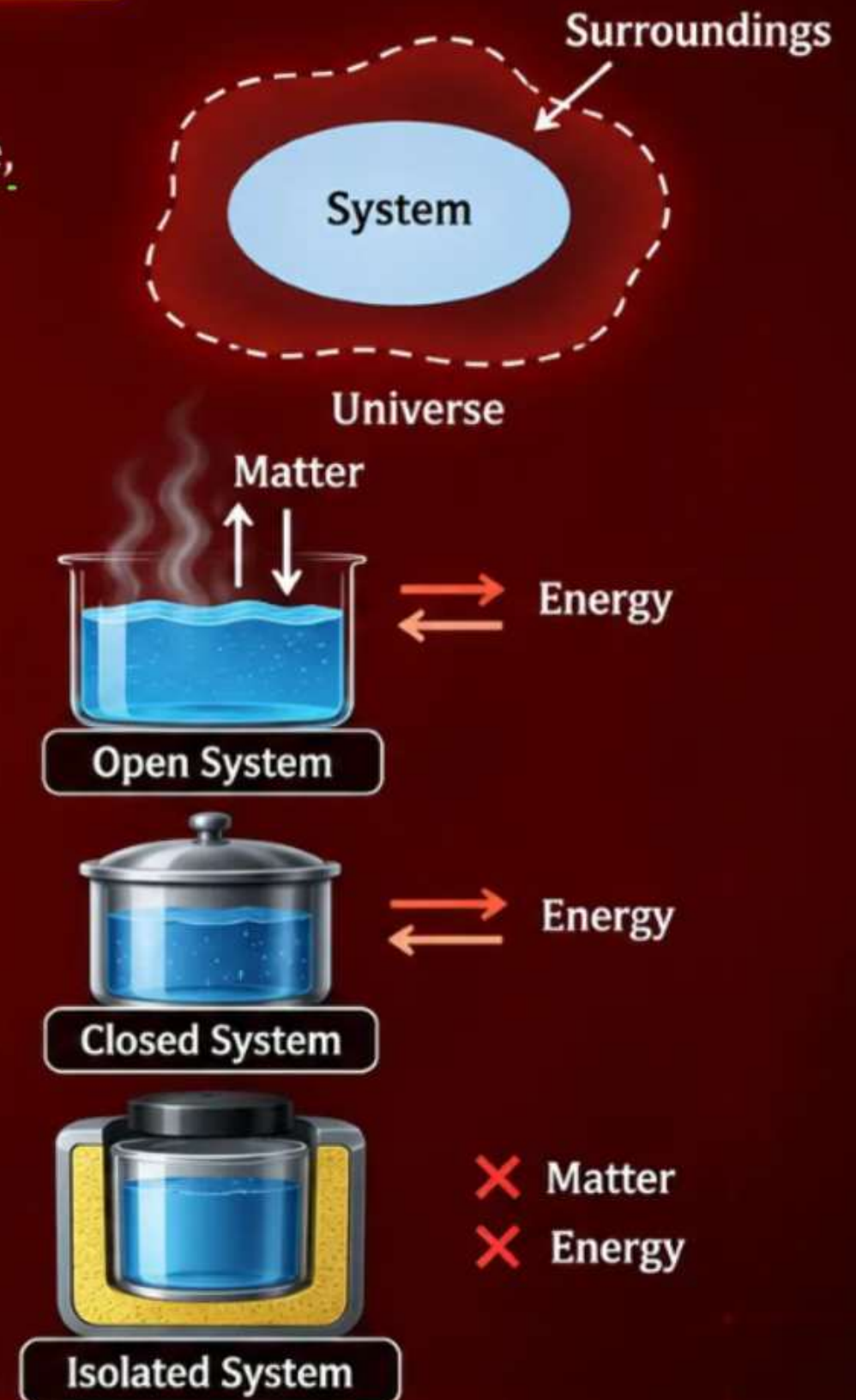
- Exchanges both matter and energy with surroundings.
- Example: Water in an open vessel.

(b) Closed System:

- Exchanges only energy, not matter.
- Example: Water in a closed metallic vessel.

(c) Isolated System:

- No exchange of matter or energy.
- Example: Water in a perfectly insulated, closed vessel.

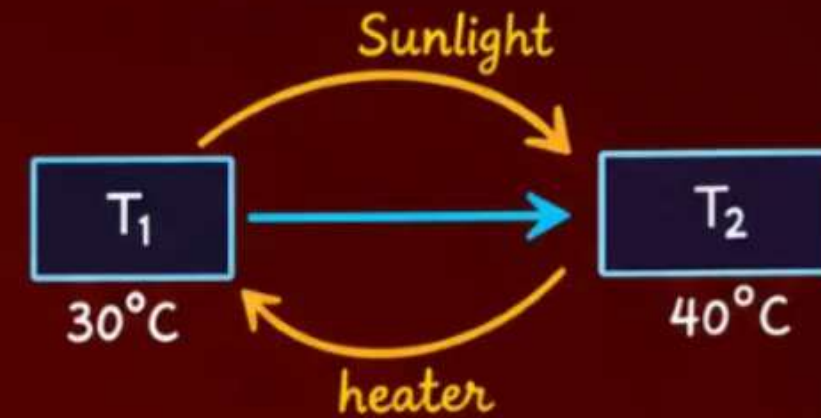


Some Basic Terms of Thermodynamics



State of the System

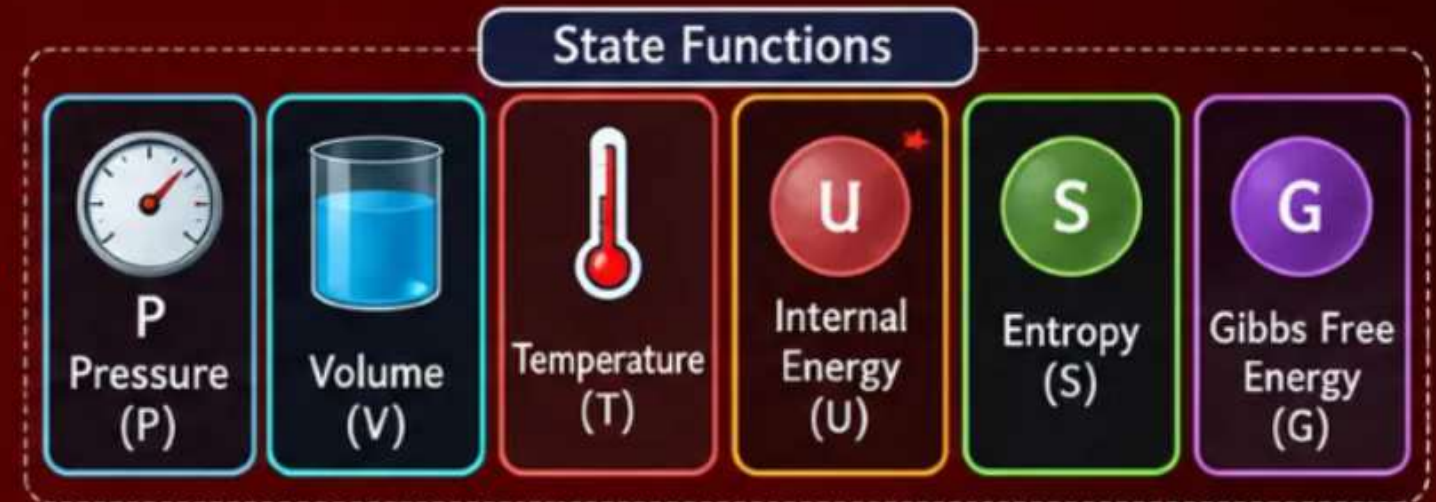
- It is the condition in which a system exists.
- This condition is expressed in terms of pressure (P), volume (V), and temperature (T).



$$\begin{aligned}\Delta T &= T_2 - T_1 \\ &= 40 - 30 \\ &= 10^\circ\text{C}\end{aligned}$$

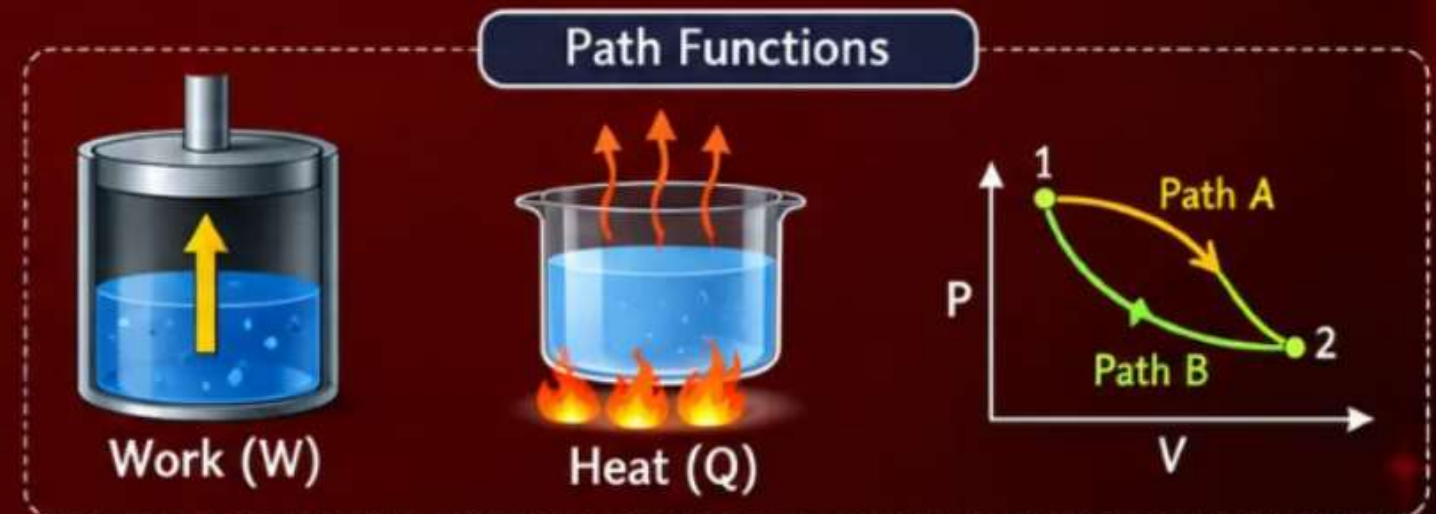
State Function

- The properties of a system that depend only on the initial and final states of the system and not on the path followed are called state functions.
- Examples: Pressure (P), Volume (V), Temperature (T), Internal Energy (U), Entropy (S), Gibbs Free Energy (G), etc.



Path Function

- The properties of a system that depend on the path or manner by which the change takes place are called path functions.
- Examples: Work (W), Heat (Q), Heat Capacity, etc.



Types of Processes



(a) Adiabatic Process:

- No heat exchange with surroundings ($q = 0$).
- Temperature changes due to work done.
- **Example:** Sudden compression or expansion of air in a piston.

Compression
(adiabatic)



$$q = 0$$

$$\Delta T \neq 0$$

No heat exchange
Temperature changes
Work done

(b) Isothermal Process:

- Temperature remains constant ($\Delta T = 0$).
- Heat exchange occurs to maintain constant temperature.
- **Example:** Slow expansion or compression of gas in a piston kept in a water bath.



Water bath
(constant temperature)

$$\Delta T = 0$$

$$q \neq 0$$

Temperature constant
Heat exchange occurs

(c) Isobaric Process:

- Pressure remains constant ($\Delta P = 0$).
- Heat changes the volume of gas.
- **Example:** Boiling water in an open vessel (pressure = atmospheric pressure).

Expansion
(at constant pressure)



$P = \text{constant}$

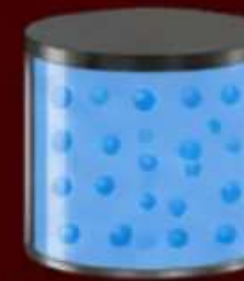
$$\Delta P = 0$$

$$q \neq 0$$

Pressure constant
Heat changes volume

(d) Isochoric Process:

- Volume remains constant ($\Delta V = 0$).
- No work is done by the system.
- **Example:** Heating a gas in a closed, rigid container.



$V = \text{constant}$

$$\Delta V = 0$$

$$q \neq 0$$

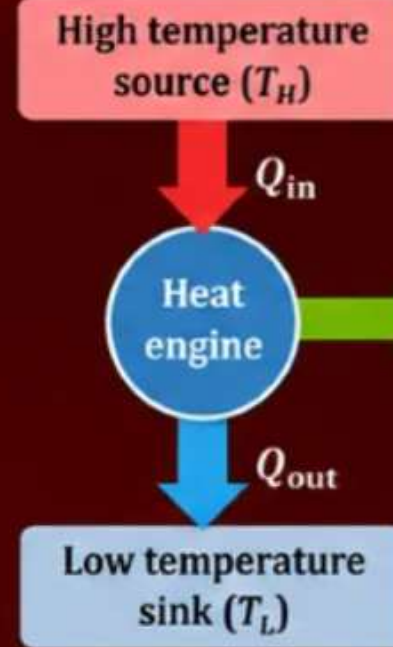
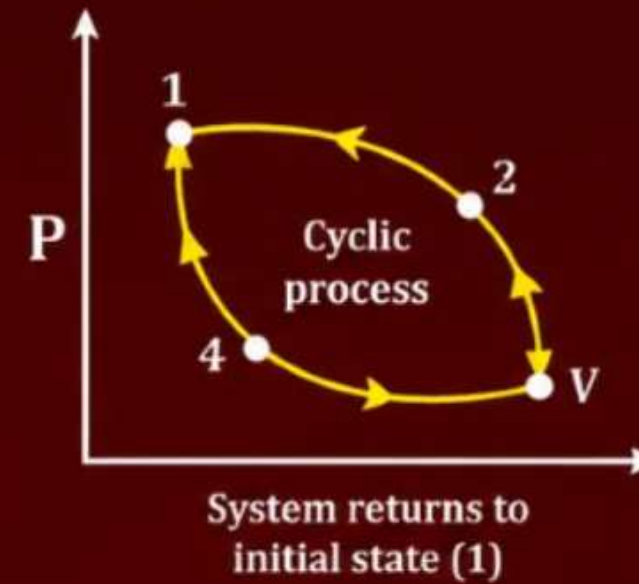
Volume constant
No work done

Types of Processes



(e) Cyclic Process:

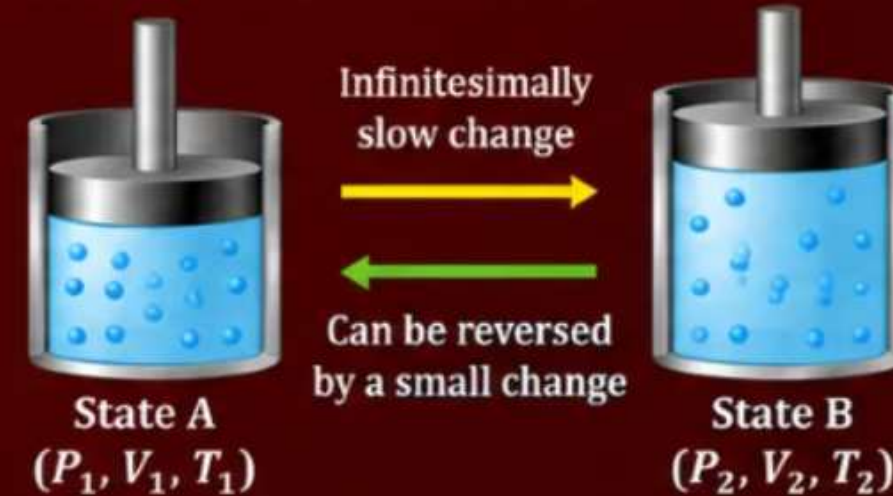
- System undergoes changes and returns to its initial state.
- Net internal energy change = 0.
- Example:** Heat engine cycle (like Carnot or Otto cycle).



$$\Delta U = 0$$

(f) Reversible Process:

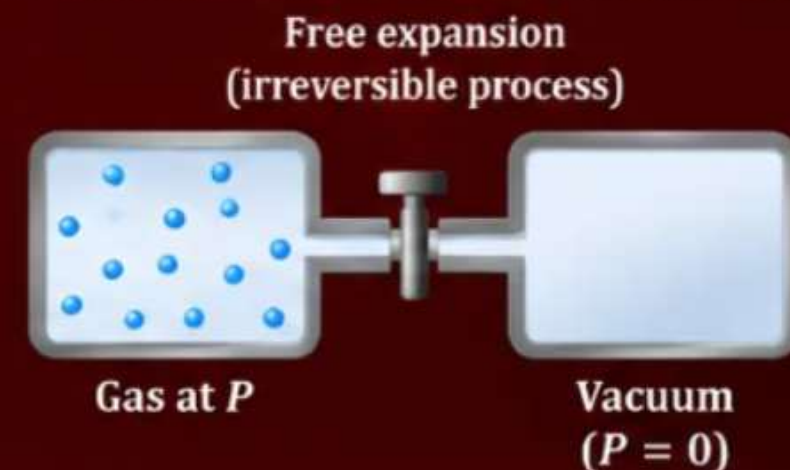
- Takes place infinitely slowly, and can be reversed by a small change.
- Gives maximum work.
- Example:** Slow, reversible expansion of gas in a piston.



Reversible
 $\Rightarrow$ Maximum work

(g) Irreversible Process:

- Occurs quickly and cannot be reversed by small changes.
- Work done is less than in a reversible process.
- Example:** Free expansion of gas into vacuum or sudden compression.



Cannot be reversed
by small changes
 $\Rightarrow$ Work done is less

Extensive and Intensive Properties

(a) Extensive properties:

- Depend on the amount of substance present.
- Examples:** Mass, volume, surface area, internal energy, enthalpy, entropy, heat capacity, etc.

no. of moles

Smaller amount of substance



n moles

Mass = m
Volume = V
Internal energy = U
Enthalpy = H
(smaller value)

Larger amount of substance



$2n$ moles

Mass = $2m$
Volume = $2V$
Internal energy = $2U$
Enthalpy = $2H$
(larger value)

(b) Intensive properties: (Independent).

- Do not depend on the amount of substance.
- Examples:** Temperature, pressure, viscosity, surface tension, refractive index, etc., *Density*.

1 mole of water



Same intensive properties

Temperature = 25°C
Pressure = 1 atm
Viscosity = η
Surface tension = γ
Refractive index = n

2 moles of water



Same intensive properties

Temperature = 25°C
Pressure = 1 atm
Viscosity = η
Surface tension = γ
Refractive index = n

Intensive properties are independent of the amount of substance.

Internal Energy (U):

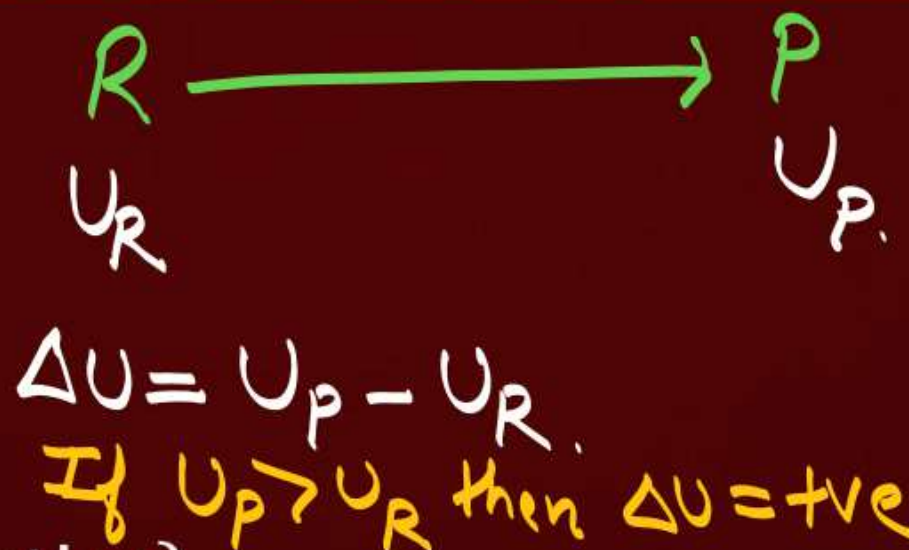
($\mathcal{E}$)



- It is the **total of all forms of energy** (like kinetic, potential, and vibrational energy) present inside a system.
- $U = \text{Kinetic energy} + \text{Potential energy} + \text{Vibrational energy} + \dots$
- In thermodynamics, we usually talk about **change in internal energy (ΔU).**

If a chemical reaction is:

- $R \rightarrow P$
- $\Delta U = U_P - U_R$
- If $U_R > U_P$, then $\Delta U = -$ (negative)



- Since ΔU depends only on initial and final states, **internal energy is a state function** and also an **extensive property**.

Modes of Transference of Energy between System and Surroundings

1. Heat (q):



Heat is the energy transferred between a system and its surroundings due to a temperature difference.

- Heat flows from high temperature to low temperature.
- **q is positive** → when heat enters the system (absorbed).
- **q is negative** → when heat leaves the system (released).

2. Mechanical or Pressure-Volume Work:

- This is the work done by gases when they expand or contract against an external pressure (like atmospheric pressure).
- It is the most common type of work in thermodynamics.

Suppose a cylinder containing is compressed with the help of piston V_1 to V_2 .

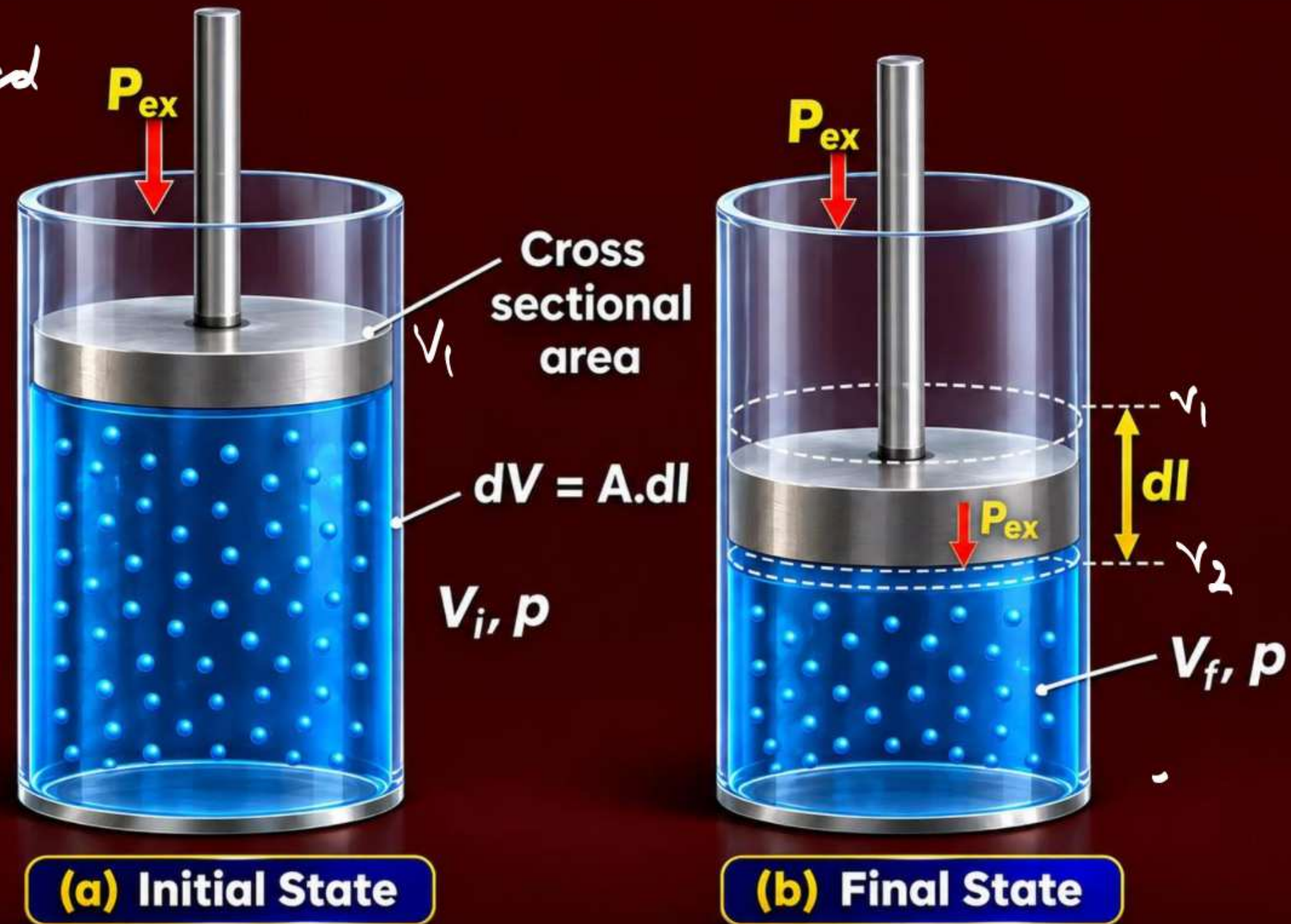


Fig. Pressure-Volume work

a) Calculation of Irreversible Work.

Let during compression the

$$\text{Volume change} = l \times A \text{ or } \Delta V = V_2 - V_1$$

$$\therefore \text{Pressure} = \frac{\text{Force}}{\text{Area}}$$

$$\text{or Force} = P_{\text{ext}} \times \text{Area}$$

$$\begin{aligned} \text{Work done} &= \text{Force} \times \text{displacement} \\ &= P_{\text{ext}} \times A \times l \end{aligned}$$

$$W_{\text{rev}} = P_{\text{ext}} \Delta V.$$

$$W_{\text{rev}} = - P_{\text{ext}} \Delta V$$

b) Calculation of Reversible work.



We know that, the process completes in infinite step, so have to calculate work of each step, by integrating.

$$\int \delta W_{\text{rev}} = - \int_{V_1}^{V_2} P_{\text{ext}} dV.$$

$$W_{\text{rev}} = - \int_{V_1}^{V_2} P_{\text{ext}} dV.$$

For compression, $P_{\text{ext}} > P_{\text{int}}$ i.e. $P_{\text{ext}} = P_{\text{int}} + dP$

For expansion, $P_{\text{ext}} < P_{\text{int}}$ i.e. $P_{\text{ext}} = P_{\text{int}} - dP$

i.e. $P_{\text{ext}} = \underline{P_{\text{int}} \pm dP}$

$$W_{\text{rev}} = - \int_{V_1}^{V_2} (P_{\text{int}} \pm dP) dV.$$

$$W_{rev} = - \int_{V_1}^{V_2} (P_{ext} \pm dp) dV.$$

$$= - \int_{V_1}^{V_2} P_{ext} dV \pm \underline{dp} \underline{dV}.$$

$$W_{rev} = - \int_{V_1}^{V_2} P_{ext} dV.$$

$\left\{ \begin{array}{l} dp dV = \text{small change} \\ \text{can be neglected} \end{array} \right\}$

According to ideal gas equation $PV = nRT$,
or $P = \frac{nRT}{V}$.

$$= - \int_{V_1}^{V_2} \frac{nRT}{V} dV.$$

$$W_{rev} = -nRT \int_{V_1}^{V_2} \frac{dV}{V}$$

$$W_{rev} = -nRT \int_{V_1}^{V_2} \frac{dV}{V}.$$

$$W_{rev} = -nRT \ln \frac{V_2}{V_1}$$

$$W_{rev} = -2.303 nRT \log \frac{V_2}{V_1}$$

$$\left[\begin{array}{l} \int_{x_1}^{x_2} \frac{dx}{x} = \ln \frac{x_2}{x_1} \\ \therefore \ln = 2.303 \log \end{array} \right]$$

Sign Convention for Work: $(V_2 > V_1)$ ✓

i) Work done on the system (Compression) = +ve.

$$\begin{aligned} W &= -P_{ext} (V_2 - V_1). \quad V_1 > V_2 \\ &= -ve (-ve) \\ &= \underline{+ve}. \end{aligned}$$

ii) Work done by the system (Expansion) = -ve.

$$\begin{aligned} W &= -P_{ext} (V_2 - V_1) \quad (V_2 > V_1) \\ &= -ve (+ve) \\ &= \underline{-ve}. \end{aligned}$$



Work of expansion in Vacuum. ($P_{\text{ext}} = 0$)

$$W = - P_{\text{ext}} \Delta V$$

$$W = 0$$

First Law of Thermodynamics

(Most Imp.)



- The first law of thermodynamics is simply the law of conservation of energy which states that Energy can neither be created nor destroyed although it may be converted from one form to another.
- Or
- The total energy of the universe (i.e., the system and the surroundings) remains constant, although it may undergo transformation from one form to the other.

Example:

- (a) **Photosynthesis** → Light energy is converted into chemical energy.
- (b) **Burning wood** → Chemical energy is converted into light and heat energy.
- (c) **Electric bulb** → Electrical energy is converted into light energy.

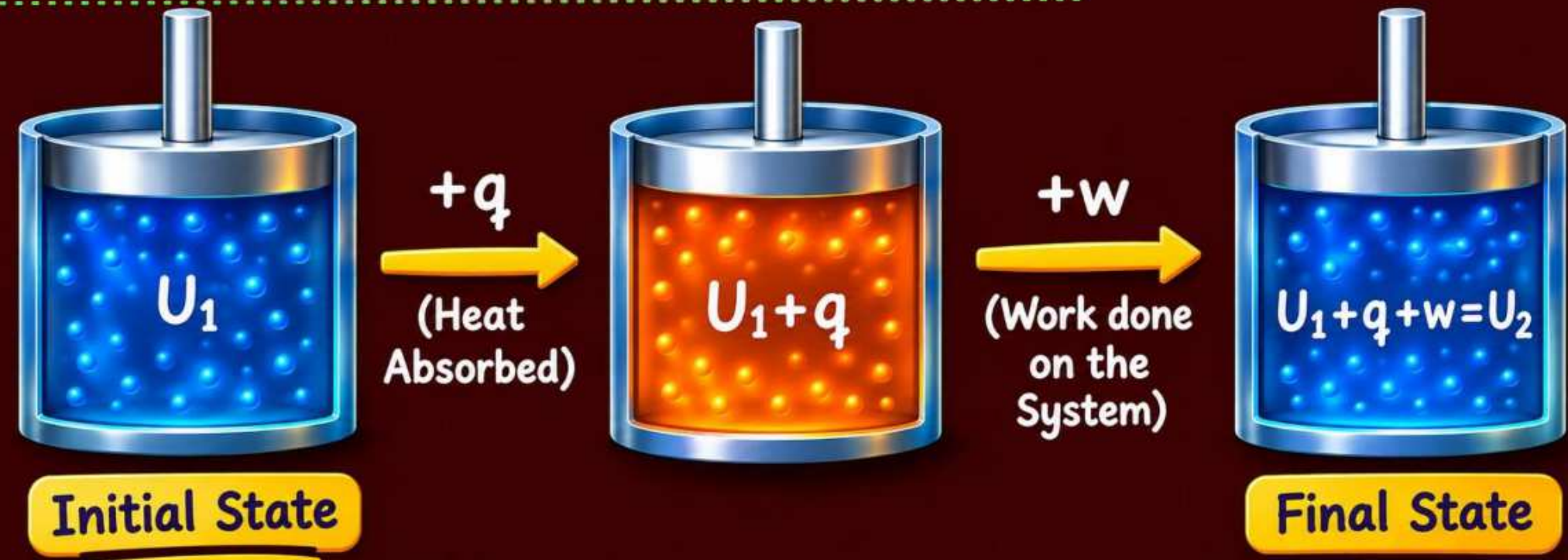
First Law of Thermodynamics



Mathematical derivation

Let the system at its initial state has internal energy U_1

Internal energy
can be changed
by transferring
heat & by
doing work.



Let the Internal energy of Final state

$$U_2 = U_1 + q + w$$

$$U_2 - U_1 = q + w$$

$$\Delta U = q + w$$

OR

$$\Delta U = q + (-P\Delta V)$$

$$\Delta U = q - P\Delta V$$

$$q = \Delta U + P\Delta V$$

Enthalpy

(H)



- The energy stored within the substance or the system that is available for conversion into heat is called the heat content or enthalpy of the substance or the system. ~~mathematically~~ mathematically by the equation.

$$H = U + PV$$

$$H = U + PV$$

- Heat absorbed or released at constant P.

We know that, FLOT.

$$q_p = \Delta U + P\Delta V. \quad - \quad 1$$

$$q_p = (U_2 - U_1) + P(V_2 - V_1).$$

$$q_p = U_2 - U_1 + PV_2 - PV_1$$

$$q_p = U_2 + PV_2 - U_1 - PV_1$$

$$q_p = (U_2 + PV_2) - (U_1 + PV_1)$$

$$q_p = H_2 - H_1$$

★

$$q_p = \Delta H$$

Put in equation ①

$$\Delta H = \Delta U + P\Delta V$$

Heat absorbed or released at constant volume



According to FLOT

$$q_v = \Delta U + P\Delta V$$

But Volume = constant, $\Delta V = 0$.

$$q_v = \Delta U$$

$$q_p = \Delta H$$

$$q_v = \Delta U$$

Relation Between q_p & q_v

We know that

$$H = U + PV.$$

or $\Delta H = \Delta U + \Delta PV.$

$$\Delta H = \Delta U + (P_2V_2 - P_1V_1)$$

According to ideal gas equation

$$PV = nRT$$

i.e. $P_1V_1 = n_1RT.$

$$P_2V_2 = n_2RT.$$

$$\Delta H = \Delta U + n_2RT - n_1RT$$

$$\Delta H = \Delta U + (n_2 - n_1)RT$$

*

$$\Delta H = \Delta U + \Delta n_g RT$$

$$q_p = q_v + \Delta n_g RT.$$

$n_2 - n_1 = \Delta n_g$
 $\downarrow$
 change in no. of gaseous moles during Reaction.



$$\begin{aligned} \Delta n_g &= n_2 - n_1 = n_p - n_R \\ &= 2 - 4 \\ &= \boxed{-2} \end{aligned}$$

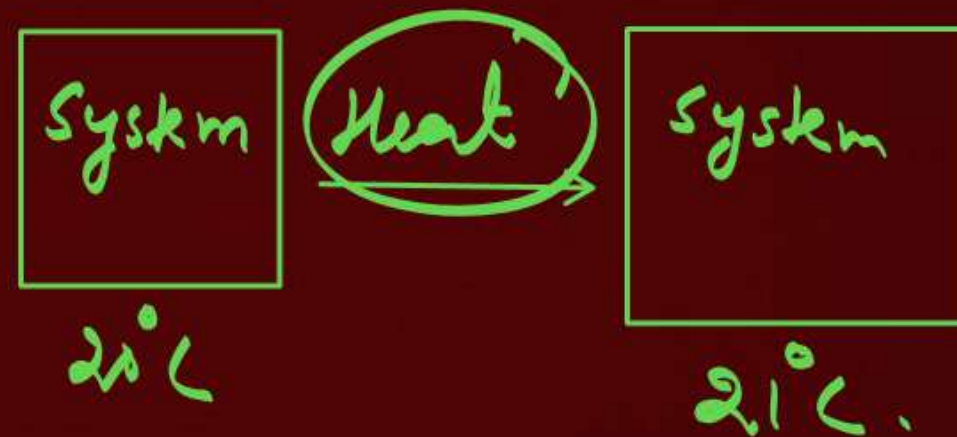
Heat Capacity, Specific Heat Capacity and Molar Heat Capacity

- It is defined as the amount of heat required to raise the temperature of a system by one degree (Celsius or Kelvin).
- It is denoted by C .

Mathematically:

$$C = \frac{q}{\Delta T}$$

- It is a **path function**.
- It is an **extensive property**.



Heat Capacity, Specific Heat Capacity and Molar Heat Capacity

Specific Heat Capacity:

- It is defined as the amount of heat required to raise the temperature of **1 g of a substance** by one degree (Celsius or Kelvin).
- It is denoted by C_s .

Mathematically:

$$C_s = \frac{q}{\Delta T \times m} \quad \text{or} \quad C_s = \frac{C}{m}$$

- It is a **function with state**
- It is an **intensive property**.



Heat Capacity, Specific Heat Capacity and Molar Heat Capacity

Molar Heat Capacity

- It is defined as the amount of heat required to raise the temperature of **1 mole of a substance** by one degree (Celsius or Kelvin).
- It is denoted by C_m .

Mathematically:

$$C_m = \frac{q}{\Delta T \times n} \quad \text{or} \quad C_m = \frac{C}{n}$$

- It is a **function ~~with~~ state**.
- It is an **intensive property**.



Relation Between C_p & C_v

Heat Capacity at Constant Pressure (C_p)

$$C_p = \frac{q_p}{\Delta T}$$

or $C_p \Delta T = q_p$ or $q_p = C_p \Delta T = \Delta H$

Heat Capacity at Constant Volume (C_v)

$$C_v = \frac{q_v}{\Delta T}$$

or $C_v \Delta T = q_v$ or $q_v = C_v \Delta T = \Delta U$

We know that

$$H = U + PV.$$

$$\Delta H = \Delta U + \Delta PV.$$

According to ideal gas equation for one mole of ideal gas $PV = RT$.

$$\Delta H = \Delta U + \Delta RT.$$

$$\Delta H = \Delta U + R\Delta T.$$

$$C_p \Delta T = C_v \Delta T + R\Delta T.$$

Dividing b/s by ΔT , we get

$$C_p \frac{\cancel{\Delta T}}{\cancel{\Delta T}} = C_v \frac{\cancel{\Delta T}}{\cancel{\Delta T}} + R \frac{\cancel{\Delta T}}{\cancel{\Delta T}}.$$

$$C_p = C_v + R. \text{ or }$$

$$C_p - C_v = R.$$

$$C_p - C_v = nR.$$

For n -mole of ideal gas.

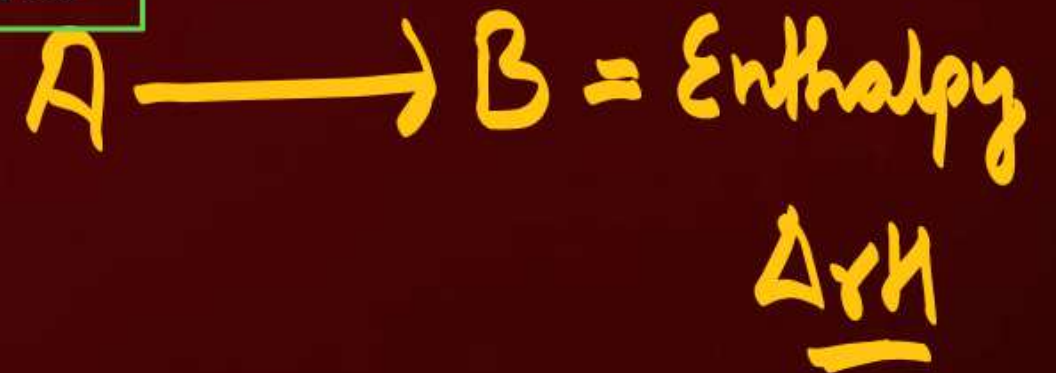
Relation Enthalpy ($\Delta_r H$)

It is defined as the enthalpy change when a given number of moles of reactants react completely to form products.

- It can be positive (+ve) → Endothermic reaction (heat absorbed). ($\Delta_r H = +ve$)
- It can be negative (-ve) → Exothermic reaction (heat released). ($\Delta_r H = -ve$).

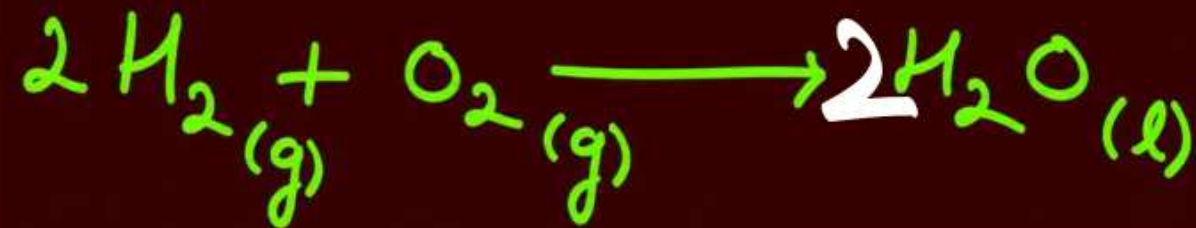
Formula:

$$\Delta_r H = \Sigma \text{ Enthalpy of products} - \Sigma \text{ Enthalpy of reactants}$$



Standard Enthalpy of Reaction ($\Delta_r H^\circ$)

- It is defined as the enthalpy change when the number of moles of reactants react completely to form products, with all reactants and products in their standard states (most stable form at 298 K and 1 bar pressure).
- It is represented by $\Delta_r H^\circ$.

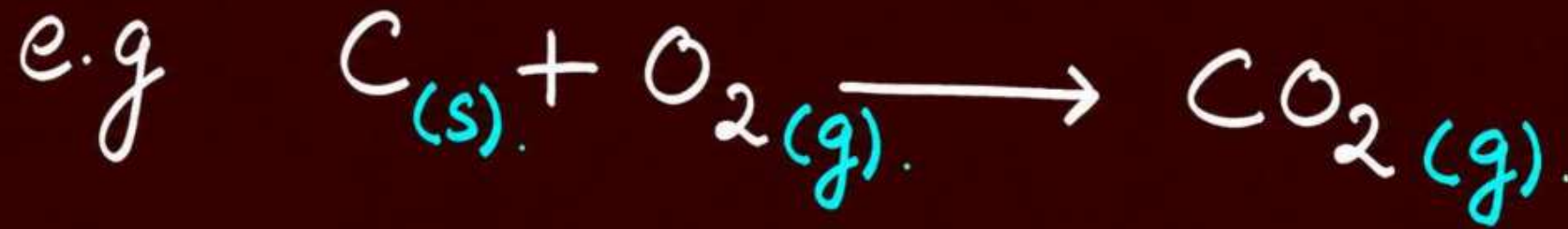


$$\Delta_r H^\circ = \text{---} \text{ kJ mol}^{-1}$$

Standard Enthalpy of Formation ($\Delta_f H^\circ$)



- It is the enthalpy change when one mole of a compound is formed from its constituent elements in their standard states (most stable form at 298 K and 1 bar).



$$\Delta_f H^\circ = \text{X kJ mol}^{-1}$$



↓
2mol

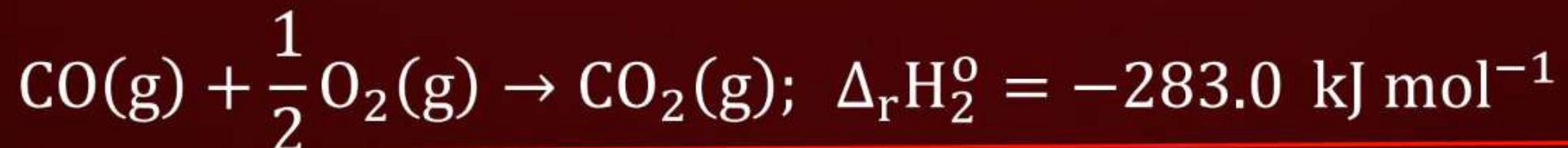
$$\Delta_f H^\circ = \text{X}$$

$\Delta_f H^\circ$ for standard states of elements is taken as zero.

Hess's Law of Constant Heat Summation

- It states that the enthalpy change (or heat evolved or absorbed) in a particular reaction is the same whether the reaction takes place in one step or in a number of steps.

Illustration of Hess's Law. Let us illustrate the law by considering the formation of carbon dioxide from carbon and oxygen. Carbon can be converted into CO_2 by two ways:



Applications of Hess's Law

1. Enthalpy of formation:

- Used to find the enthalpy of formation of compounds that cannot be measured directly, like CH_4 , CO , and benzene.

2. Enthalpy of transition:

- Helps calculate the enthalpy change when one allotropic form changes to another, e.g., graphite diamond, rhombic sulphur $\rightarrow$ monoclinic sulphur.

Graphite $\rightarrow$ Diamond

3. Enthalpy of hydration:

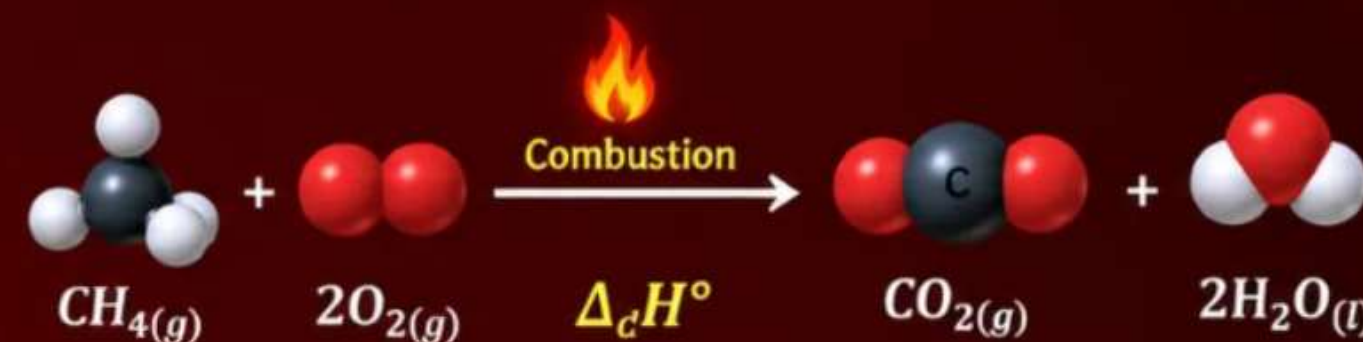
- Used to find the enthalpy change when anhydrous salts combine with water to form hydrated salts.

Different Types of Enthalpies



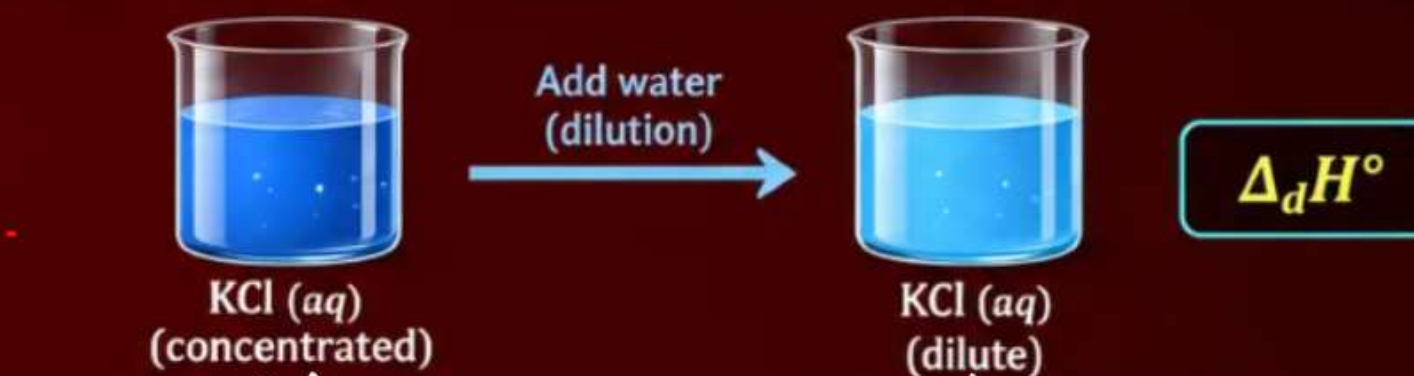
1. Standard Enthalpy of Combustion ($\Delta_c H^\circ$):

- It is defined as the enthalpy change when one mole of a substance is completely burnt in the presence of oxygen, provided all reactants and products are in their standard states.



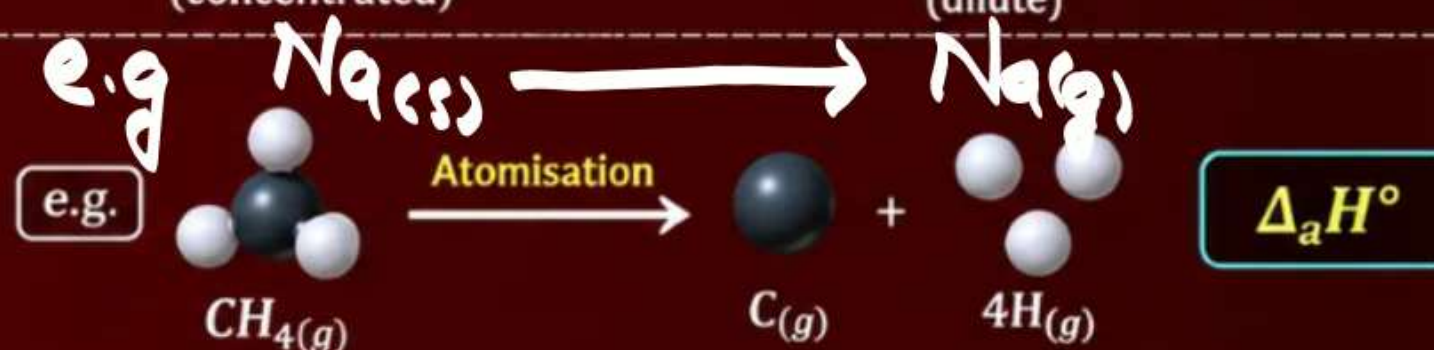
2. Enthalpy of Dilution:

- It is defined as the enthalpy change when one mole of a substance is diluted from one concentration to another.



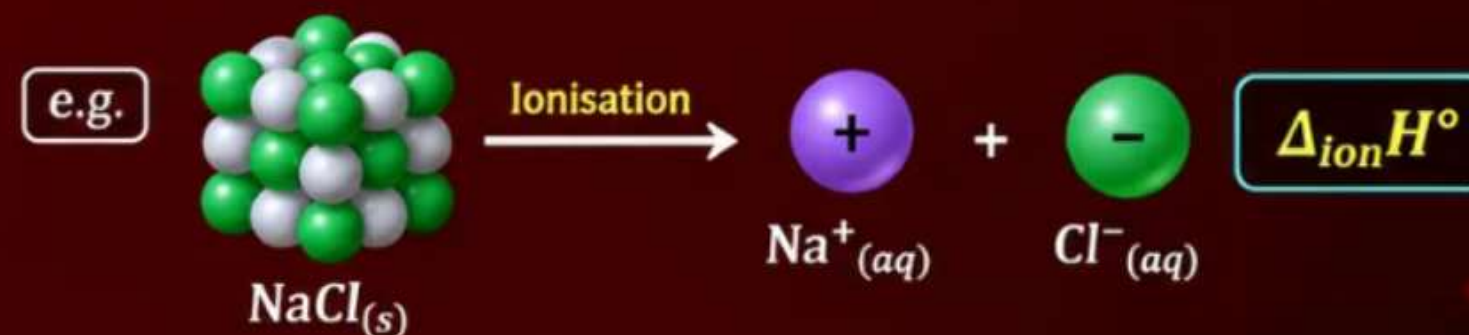
3. Enthalpy of Atomisation ($\Delta_a H^\circ$):

- It is the change in enthalpy when one mole of a substance is completely dissociated into its gaseous atoms.



4. Enthalpy of Ionisation ($\Delta_{\text{ion}} H^\circ$):

- It is the change in enthalpy when one mole of an electrolyte is completely dissociated into its ions.



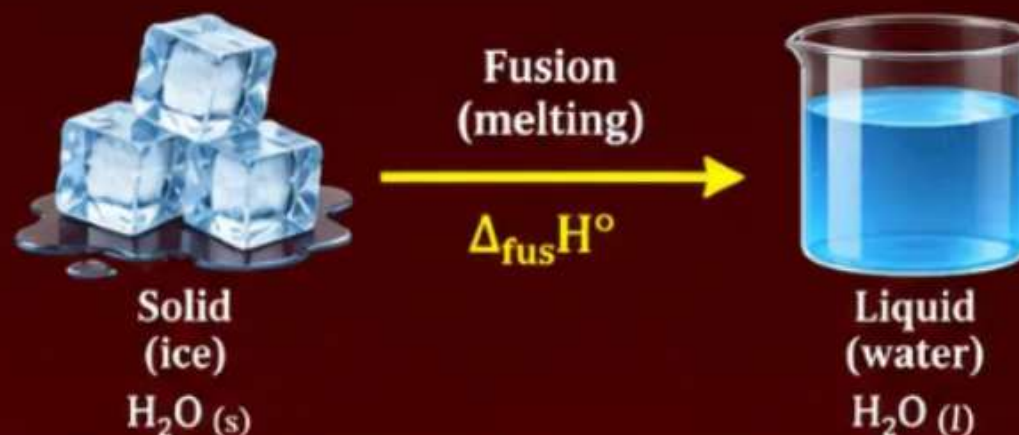
Enthalpy Change During Phase Transformations



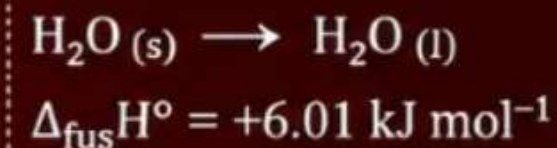
(a) Enthalpy of Fusion ($\Delta_{\text{fus}}H^\circ$):

- It is defined as the enthalpy change when one mole of a solid substance changes into liquid.

$$\Delta_{\text{fus}}H^\circ > 0 \text{ (Endothermic)}$$



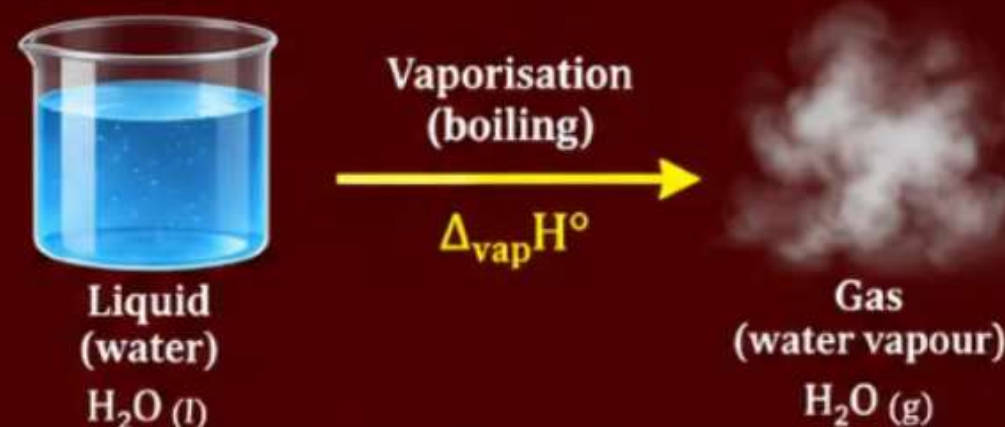
Example:



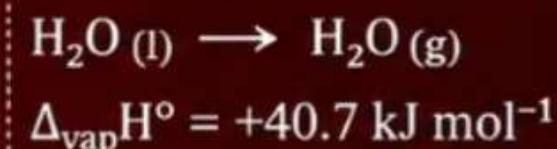
(b) Enthalpy of Vaporisation ($\Delta_{\text{vap}}H^\circ$):

- It is defined as the enthalpy change when one mole of a liquid substance changes into gaseous state.

$$\Delta_{\text{vap}}H^\circ > 0 \text{ (Endothermic)}$$



Example:



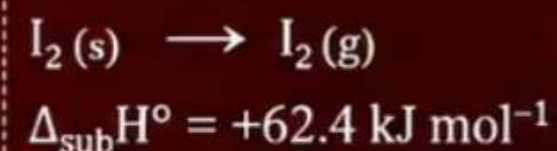
(c) Enthalpy of Sublimation ($\Delta_{\text{sub}}H^\circ$):

- It is defined as the enthalpy change when one mole of a substance changes directly from solid state to gaseous state.

$$\Delta_{\text{sub}}H^\circ > 0 \text{ (Endothermic)}$$



Example:



Sublimation is the sum of fusion and vaporisation enthalpies.

$$\Delta_{\text{sub}}H^\circ = \Delta_{\text{fus}}H^\circ + \Delta_{\text{vap}}H^\circ$$

Spontaneous Process

- A change or process that take place itself or require little initiation.
- *flow* Of heat from hotter body to a colder body.
- Mixing of different gases.
- Flow of water down a hill.

Burning of Candle. (*Initiation*)

Non-Spontaneous Process

A change or process that require continuous supply of energy.

Examples of non-spontaneous processes

- (i) Flow of heat from a cold body to a hot body.
- (iii) Flow of water from ground to overhead tank.
- (v) Decomposition of water into hydrogen and oxygen.
- (iv) Diffusion of a gas from low pressure to high pressure
- (vi) Hydrolysis of sodium chloride.

Entropy



- Entropy is the measure of disorder or randomness of a system.
- The more disordered the system, the greater is its entropy.
- Entropy is an extensive property and a state function.

Order of entropy:

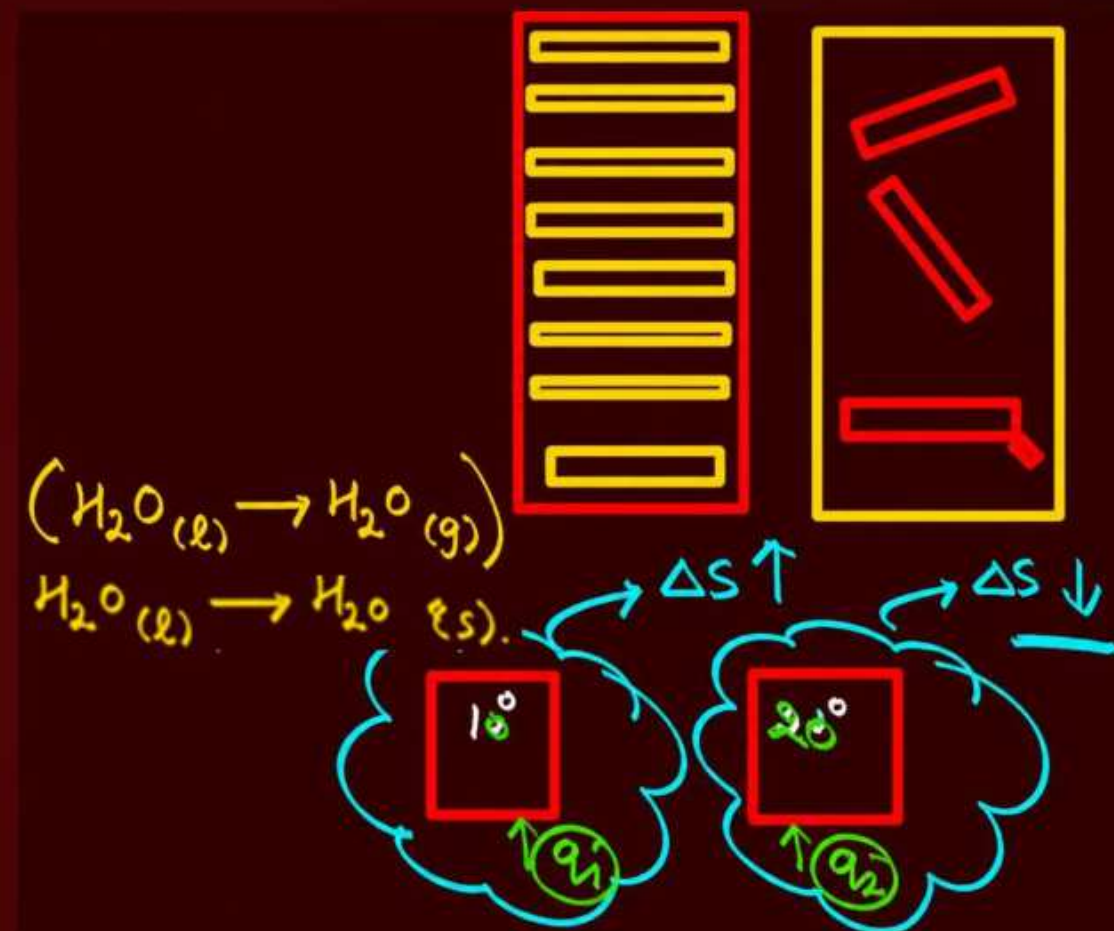
- Solid < Liquid < Gas
- In case of vaporisation or melting, entropy increases.
- In case of freezing, entropy decreases.

For a reaction:

- $A \rightarrow B$
- $\Delta S = S(\text{product}) - S(\text{reactant})$
- If two systems are placed at different temperatures and the same amount of heat is supplied to them, the entropy change for the system at lower temperature is greater.

Mathematically:

$$\Delta S = q / T$$



Entropy Change for Reversible Process



a) For Isothermal Process,
Let heat is absorbed by the
System & is released by the
surrounding, then

$$\Delta S_{\text{sys}} = \frac{q}{T} \quad \& \quad \Delta S_{\text{surr}} = -\frac{q}{T}$$

$\therefore$ Total Entropy,

$$\Delta S_{\text{univ}} = \Delta S_{\text{syst}} + \Delta S_{\text{surr}},$$

$$= +\cancel{\frac{q}{T}} - \cancel{\frac{q}{T}}$$

$$= 0$$

b) For Adiabatic Reversible process,
For adiabatic process, $q = 0$

$$\Delta S_{\text{univ}} = 0.$$

For Rev. Process

$$\Delta S_{\text{univ}} = 0.$$

Entropy Change for Reversible Process

Let us consider a Temp of a surrounding T_1 of Temp. of a system is T_2 . Heat is absorbed by the system & heat is released by the surrounding, i.e. $T_1 > T_2$, then

$$\Delta S_{\text{sys}} = +\frac{q}{T_2} \text{ \& } \Delta S_{\text{surr}} = -\frac{q}{T_1}$$

$$\begin{aligned} \Delta S_{\text{univ}} &= \frac{q}{T_2} - \frac{q}{T_1} \\ &= q \left(\frac{1}{T_2} - \frac{1}{T_1} \right) \end{aligned}$$

$$\Delta S_{\text{univ}} = q \left(\frac{T_1 - T_2}{T_1 T_2} \right)$$

$\Delta S_{\text{univ}} = +ve$, because $T_1 > T_2$.

$\Delta S_{\text{univ}} = 0$, For Reversible equilibrium
★ $\Delta S_{\text{univ}} = +ve$, For Irreversible or Natural process.

- (i) If ΔS_{total} is positive, the process is **spontaneous**.
- (ii) If ΔS_{total} is negative, the forward process is **non-spontaneous**; the reverse process may be spontaneous.
- (iii) If ΔS_{total} is zero, the process is in **equilibrium**.

$$\begin{aligned}
 \Delta S_T = +ve &\rightarrow \text{SPON} \\
 = -ve &\rightarrow \text{Non-spont} \\
 = 0 &\rightarrow \text{Equilibrium.}
 \end{aligned}$$

Gibbs Free Energy



- **Gibbs energy** of a system is defined as the maximum amount of energy available to a system during a process that can be converted into **useful work**.
- It is usually denoted by 'G' and is defined mathematically by the equation

$$G = H - TS$$

Suppose for a isothermal process,

$$G_1 = H_1 - TS_1 \text{ for initial state } \text{---} \textcircled{1}$$

$$G_2 = H_2 - TS_2 \text{ for Final state. } \text{---} \textcircled{2}$$

Subtracting $\textcircled{1}$ from $\textcircled{2}$

$$G_2 - G_1 = H_2 - TS_2 - (H_1 - TS_1)$$

$$= H_2 - TS_2 - H_1 + TS_1$$

$$\begin{aligned} \Delta G &= H_2 - H_1 - TS_2 + TS_1 \\ &= H_2 - H_1 - (TS_2 - TS_1) \end{aligned}$$

$$\Delta G = \Delta H - T(S_2 - S_1)$$

$$\Delta G = \Delta H - T\Delta S$$

Gibbs Helmholtz equation.

Criteria for Spontaneity in terms of Gibbs free Energy

Consider a isothermal process, system absorbs heat & surrounding releases heat at constant pressure. then.

$$-q_{\text{surr}} = +q_{\text{sys}} = \Delta H_{\text{sys}}$$

$$\therefore -q_{\text{surr}} = \Delta H_{\text{sys}} \quad \text{or} \quad q_{\text{surr}} = -\Delta H_{\text{sys}}$$

$$\begin{aligned} \therefore \Delta S_{\text{univ}} &= \Delta S_{\text{sys}} + \Delta S_{\text{surr}} \\ &= \Delta S_{\text{sys}} + \frac{q_{\text{sys}}}{T} \end{aligned} \quad \Delta S = \frac{q_{\text{rev}}}{T}$$

$$\Delta S_{\text{univ}} = \Delta S_{\text{sys}} + \frac{\Delta H_{\text{sys}}}{T}$$

Multiplying b/s by T.

$$T\Delta S_{\text{univ}} = T\Delta S_{\text{sys}} - \Delta H_{\text{sys}}$$

$$T\Delta S_{\text{univ}} = -\Delta H_{\text{sys}} + T\Delta S_{\text{sys}}$$

$$T\Delta S_{\text{univ}} = -(\Delta H_{\text{sys}} - T\Delta S_{\text{sys}})$$

$$-T\Delta S_{\text{univ}} = \Delta G$$

$$\Delta G = -T\Delta S_{\text{univ}}$$

ΔG = +ve, $\Delta S = -ve$ ie For non spontaneous process.

$\Delta G = -ve$, $\Delta S = +ve$, ie For spontaneous process.

$\Delta G = 0$, $\Delta S = 0$, ie For equilibrium

MORE THAN ENOUGH FOR
BOARDS.

QUESTION



Which of the following is an intensive property?

[2024]

1 Heat capacity

2 Mass

3 Density

4 Volume

QUESTION



What is an isothermal process?

[2024]

QUESTION



State first law of Thermodynamics. Derive its mathematical relation.

[2024]

QUESTION



Define entropy and free energy. How are they related to each other?

[2024]

QUESTION



Which of the following is an intensive property?

[2024]

- 1 Enthalpy
- 2 Entropy
- 3 Internal energy
- 4 Specific heat

QUESTION



What is an isobaric process?

[2024]

QUESTION



What do you understand by open and isolated systems? Give examples.

[2025]

QUESTION



State first law of thermodynamics and give a mathematical expression for it.
Mention two examples to support the law. **[2025]**

QUESTION



What are intensive and extensive properties? Give two examples of each.

[2025]

QUESTION



Do you understand by open and isolated systems ? Give examples.

[2025]

QUESTION



Which of the following is an intensive property?

[2025]

- 1** Heat capacity
- 2** Mass
- 3** Density
- 4** Volume

QUESTION



State first law of thermodynamics and give a mathematical expression for it. Mention two examples to support the law. **[2025]**

QUESTION



Define entropy and free energy. How are they related to each other?

[2025]

QUESTION



Internal energy of a substance/system is:

[2025]

- 1** State function
- 2** Path function
- 3** Neither state nor path function
- 4** Both state function as well as path function

QUESTION



Explain term enthalpy. How does it differ from internal energy?

[2025]

Thank
You